

Plant biotechnology

BT421

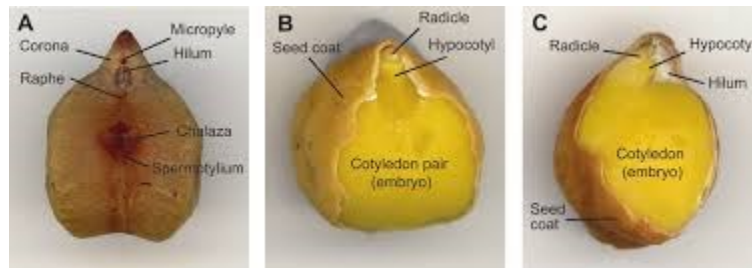
Surface sterilization

Chick pea seed

Characteristic of Chick pea seed

Dicot plant

1-2 years old for vital and germinated



Germination occurs by anabolism and catabolism (metabolism process) after 2-3 days soaked in water. (don't need hormone for growth)

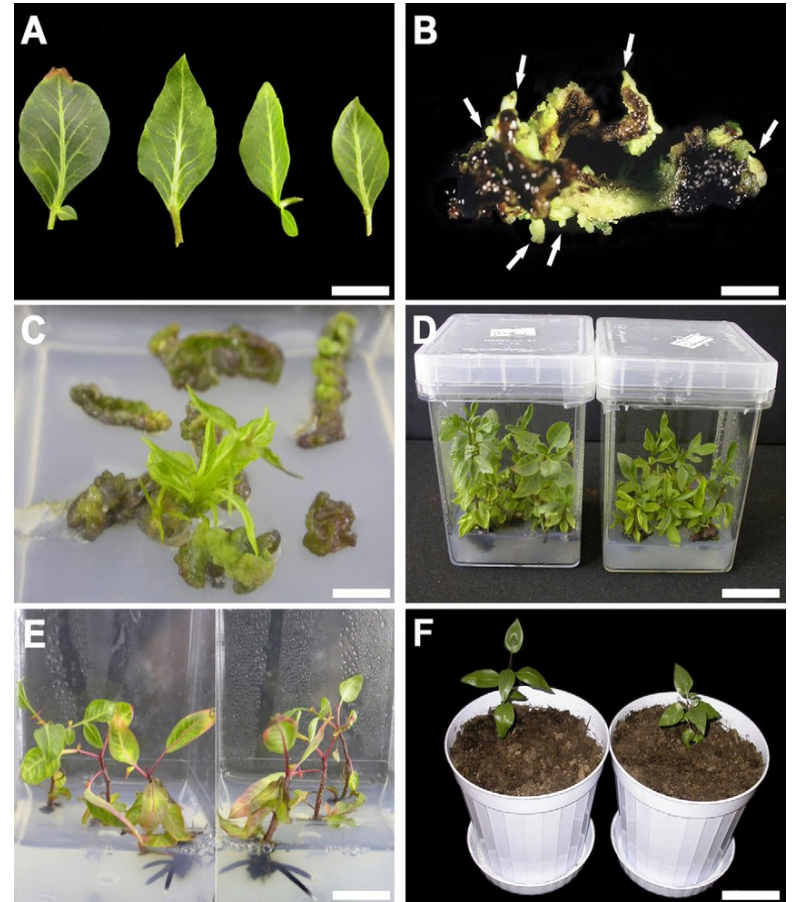


Explant (mint leaves)

Characteristic of Mint leaf

Dicot plant

Fresh leaf for vital and germinated
need hormones for germination



Method of sterilization

- 1- Methanol bromide (for soil).
- 2- Fumigation (for large field)
- 3- Radiation (UV light- Laminar flow)
- 4- Chemicals (Halogens)
- 5- Heat (Sun, Sprit lamb)
- 6- pressure (1.5 kg/cm^2)
- 7- Filtration (with heat labile growth hormone 0.2 Millipore)

Lab work

- 1- Wash seed under continues running tap water / 10min (to minimize contamination)
- 2- Wash with soap (detergent) / 10min (to break tension surface of water)
- 3- Remove extra soap (by tap water)
- 4- Wash with 70% ethanol / 30 sec.
- 5- washing with distal water .

Lab work

(work under laminar flow)

6- Wash with Disinfectant / 10 min

- Clorox (sodium hypochlorite)

- AgNO_3

- 3% H_2O_2

7- Washing with autoclaved distal water 3-4X/20min

8- Put in autoclave to the next lab .